

Data Center Impact Explorer

Pitch Presentation

Team JAMA

Anna Grace, Jillian Kunze, Andrea MacGregor, and Morgan Murphy

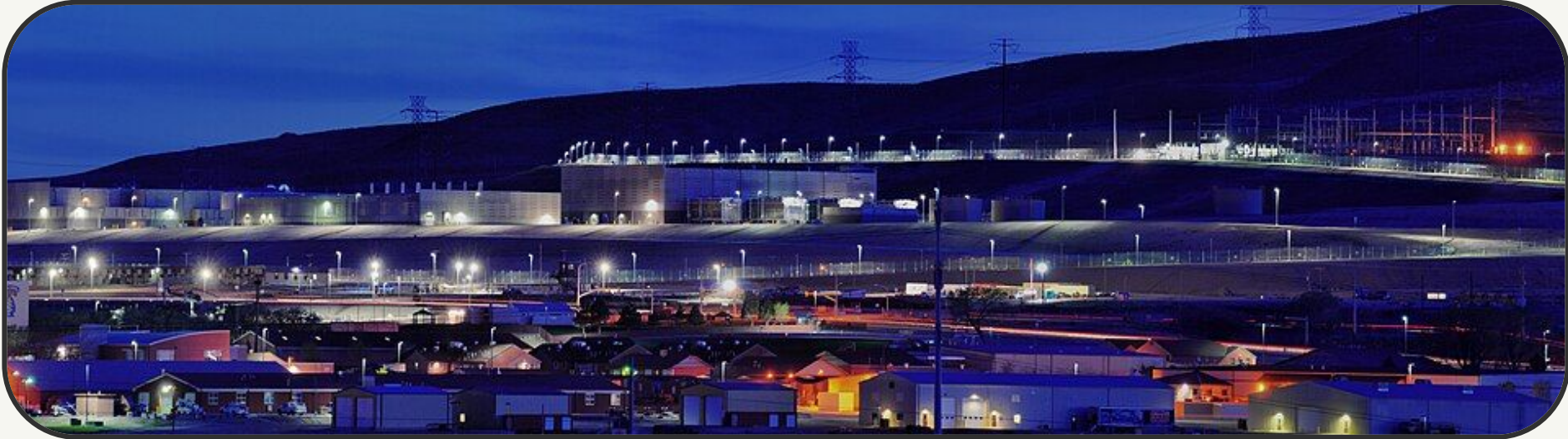


Image source: [Swilsonmc/Wikimedia Commons](#)

Introduction

- Data centers provide the physical storage space for cloud storage, digital streaming, and AI¹
 - Have long existed, but AI boom has caused dramatic growth²
- Local governments have a significant role in the trajectory of data center growth in their area based on zoning ordinances and permitting³
- Community opposition is raising concerns about the local impact of data center construction and operation,⁴ including:
 - Water usage and pollution
 - Energy usage, source, and related pollution
 - Noise pollution
- Concrete information about the aggregate impacts of data centers can be difficult to find

¹ Malone, K. (2026, February 16). A quick guide to Pennsylvania's data center debate. *Technical.ly*. <https://technical.ly/civics/pennsylvania-data-centers-explained/>

² Stansbury, M., Marchese, K., Hardin, K., & Amon, C. (2025, June 24). Can US infrastructure keep up with the AI economy? *Deloitte Insights*. <https://www.deloitte.com/us/en/insights/industry/power-and-utilities/data-center-infrastructure-artificial-intelligence.html>

³ National League of Cities & AAAS Center for Scientific Evidence in Public Issues (EPI Center). (n.d.). Community strategies to address data center development and operation.

⁴ Jones, J.M. (2026, May 13). Americans oppose AI data centers in their area. *Gallup*. <https://news.gallup.com/poll/709772/americans-oppose-data-centers-area.aspx>

Project Goals

- Analyze potential impact of data center construction on local communities based on past trends, to include:
 - Environmental Impact
 - Electricity Impact
 - Health Impact
 - Political Impact
- Consolidation of data across different sources
- Benefits the general public, specifically those who live near data centers or proposed data centers

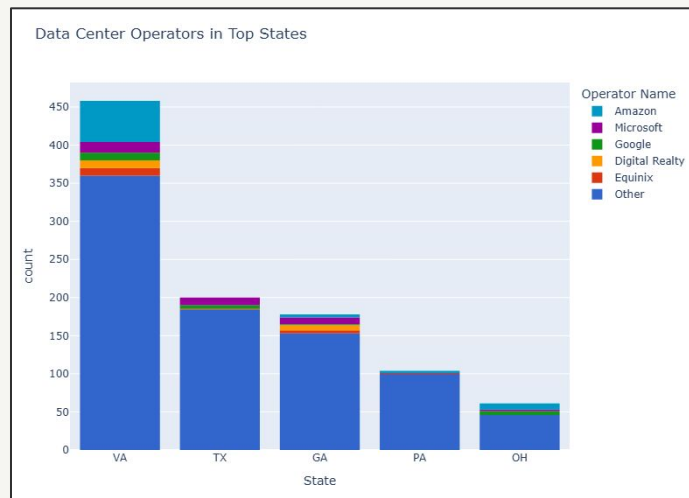
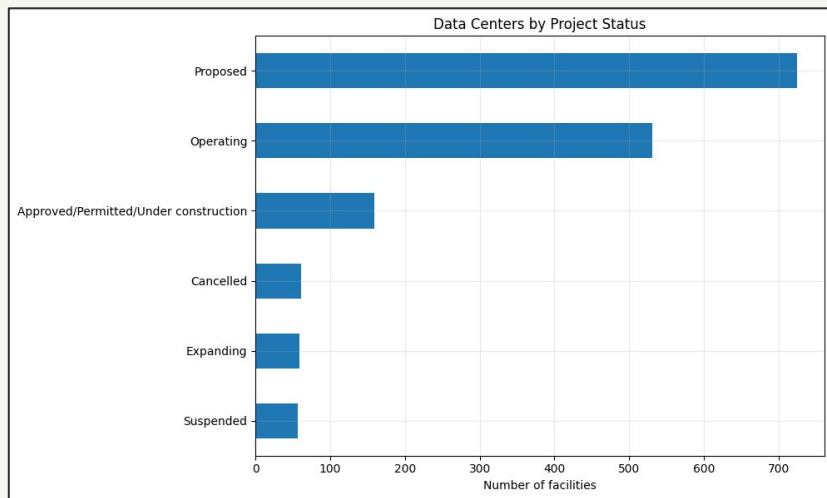
Dataset Overview

- FracTracker Alliance U.S. Data Center Database
- U.S. Energy Information Administration (EIA) Electricity Data
- America's Health Rankings (AHR)
- National Health Interview Survey (NHIS)
- U.S. Drought Monitor
- National Neighborhood Data Archive (NANDA)
- American Local Government Elections Database

Data Centers Data Overview

FracTracker Alliance U.S. Data Centers Tracker

- 1592 rows, 44 columns
- Includes location, operator (Amazon, Google, etc), status (operating, proposed, etc) and size



Environmental Aspect: Inspection

	MapDate	FIPS	County	State	None	D0	D1	D2	D3	D4	ValidStart	ValidEnd	StatisticFormatID
0	20260630	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-30	2026-07-06	1
1	20260623	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-23	2026-06-29	1
2	20260616	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-16	2026-06-22	1
3	20260609	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-09	2026-06-15	1
4	20260602	2013	Aleutians East Borough	AK	100.0	0.0	0.0	0.0	0.0	0.0	2026-06-02	2026-06-08	1

Rows: 3363768

States: 52

Counties: 1959

Earliest date: 2006-07-04 00:00:00

Latest date: 2026-07-06 00:00:00

- No null values :)
- 3,363,768 rows of data from 2006 - 2026
- 52 states - including DC and Puerto Rico

Environmental Aspect: EDA

▼ **Fig. 1** U.S. Drought trends over the past 20 years

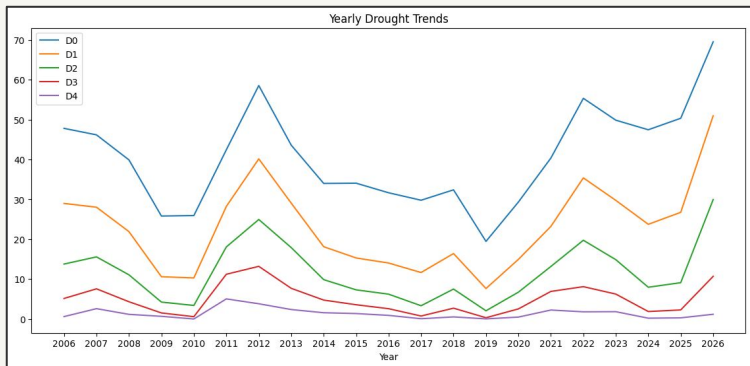
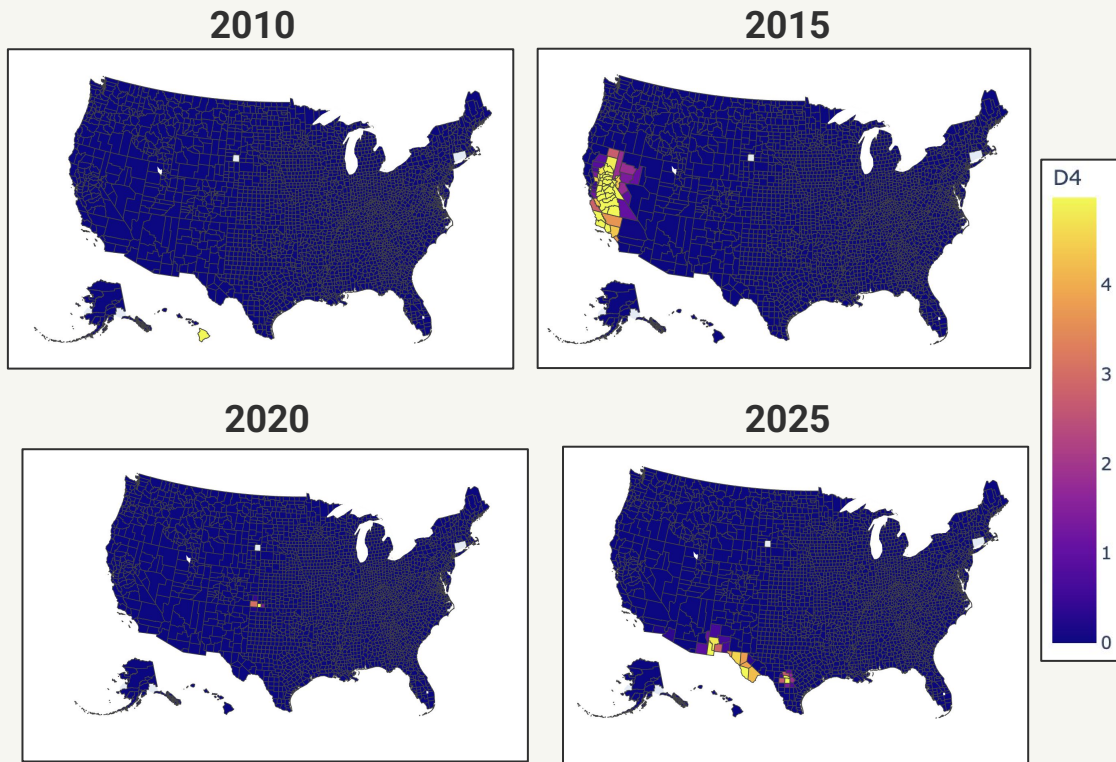


Fig. 2 U.S. D4 Drought mapped in 5 year intervals



Environmental Aspect: Next Steps

- Incorporate Data Center data into the Environmental Data
 - Map Zip Codes from Data Center data to FIP Codes from Drought data
 - Add pinpoints of data center to the maps
 - Look at number of data centers over time in tandem with drought impact over time
- Create an interactive map where years and level of drought severity can be filtered

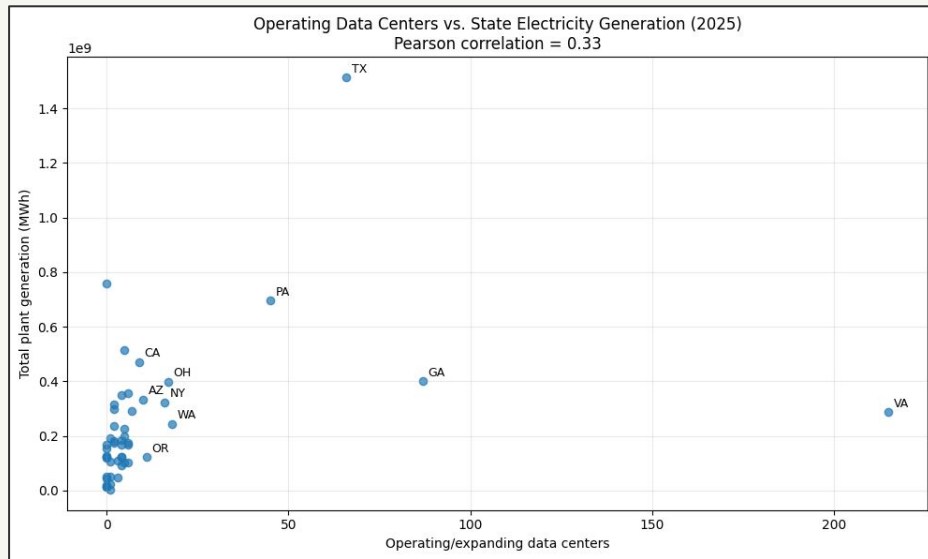
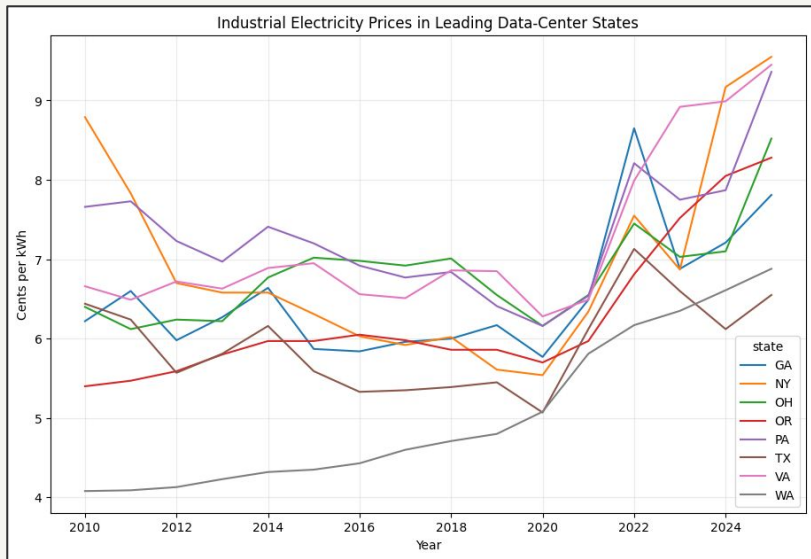
Electricity Aspect: Inspection

US Energy Information Administration Electricity Dataset

- ~8GB of JSON files concatenated into a single txt file
- Each JSON covered a single demographic:
 - Geographical Area (State)
 - Sector (Residential, commercial, industrial, etc)
 - Type (Customer, price, power generated, etc)
 - Time Frame (Annual, Quarterly, Monthly)

	dataset	rows	columns
0	data_centers	1592	44
1	customers_annual	289	26
2	customers_monthly	289	227
3	customers_quarterly	289	81
4	generation_annual	57485	33
5	generation_monthly	57550	311
6	generation_quarterly	57540	109
7	price_annual	352	33
8	price_monthly	352	311
9	price_quarterly	352	109

Electricity Aspect: EDA



Electricity Aspect: Next Steps

- Incorporate projected electricity price and generation into analysis
 - Projected data provided by EIA
- Find data on electricity with more granular geographical information like town, zipcode or fips
 - This would allow us to see the impact on a more local scale
- Creating a map of the electrical generation and pricing and incorporate data center locations

Health Aspect: Inspection

America Health Rankings: **82,054 rows, 10 columns**

- Data on value, score, and confidence interval ultimately removed prior to EDA
- Measures split into various niche demographics

	Report	Measure	State	Value	Score	Rank	LowerCI	UpperCI	Source	Data Year(s)
0	2025 Annual	Social and Economic Factors	AK	0.043	0.043	28.0	NaN	NaN	America's Health Rankings composite measure	2025
1	2025 Annual	Social and Economic Factors	AL	-0.662	-0.662	44.0	NaN	NaN	America's Health Rankings composite measure	2025
2	2025 Annual	Social and Economic Factors	ALL	NaN	NaN	NaN	NaN	NaN	America's Health Rankings composite measure	2025
3	2025 Annual	Social and Economic Factors	AR	-0.933	-0.933	48.0	NaN	NaN	America's Health Rankings composite measure	2025
4	2025 Annual	Social and Economic Factors	AZ	0.086	0.086	25.0	NaN	NaN	America's Health Rankings composite measure	2025

Report NaN percent: 0.0000%
 Measure NaN percent: 0.0000%
 State NaN percent: 0.0000%
 Value NaN percent: 15.5227%
 Score NaN percent: 94.6462%
 Rank NaN percent: 23.8172%
 LowerCI NaN percent: 22.6534%
 UpperCI NaN percent: 22.6534%
 Source NaN percent: 0.0000%
 Data Year(s) NaN percent: 0.0000%

REGION	URBRL23	HXX	DXRESP_A	HLTHCOND_A	MEDRXTRT_A	ARTHEV_A	COPDEV_A	CANEV_A	ASEV_A	SLPREST_A
0	3	4	H067658	NaN	2	2	2	2	2	3
1	3	4	H076577	NaN	2	2	2	2	2	3
2	3	4	H019335	2.0	2	2	1	1	2	4
3	3	4	H012701	2.0	2	2	1	2	2	2
4	3	2	H049678	1.0	2	2	2	2	1	3

REGION NaN percent: 0.0000%
 URBRL23 NaN percent: 0.0000%
 HXX NaN percent: 0.0000%
 DXRESP_A NaN percent: 50.5685%
 HLTHCOND_A NaN percent: 0.0000%
 MEDRXTRT_A NaN percent: 0.0000%
 COPDEV_A NaN percent: 0.0000%
 CANEV_A NaN percent: 0.0000%
 ASEV_A NaN percent: 0.0000%
 SLPREST_A NaN percent: 0.0000%
 DEPEV_A NaN percent: 0.0000%
 ANNEXV_A NaN percent: 0.0000%

National Health Interview Survey (adults):
32629 rows, 630 columns

- Detailed, granular data that covers multiple conditions and statuses
- Codebook utilized to understand naming conventions and identify useful features

Variable	Universe	Question ID	Content
SLPREST_A	HHSTAT_A=1	SLP.0020.00.2	Wake up well rested, past 30 days
ASEV_A	HHSTAT_A=1	AST.0010.00.1	Ever had asthma
CANEV_A	HHSTAT_A=1	CAN.0010.00.1	Cancer or a malignancy of any kind?
COPDEV_A	HHSTAT_A=1	CON.0010.00.1	Ever had COPD
MEDRXTRT_A	HHSTAT_A=1	ISN.0010.00.3	Weakened immune system due to prescriptions
HLTHCOND_A	HHSTAT_A=1	ISN.0020.00.3	Weakened immune system due to health condition
ANNEXV_A	HHSTAT_A=1	CON.0040.00.1	Ever had anxiety disorder
DEPEV_A	HHSTAT_A=1	CON.0050.00.1	Ever had depression
DXRESP_A	HHSTAT_A=1 and CURRESP_A=1	ALG.0030.00.2	Respiratory allergy diagnosis
HXX			Randomly assigned household number unique to a household
URBRL23			Urbanization level
REGION			Household region

Health Aspect: EDA

U.S. States Average Health Ranking for 2024

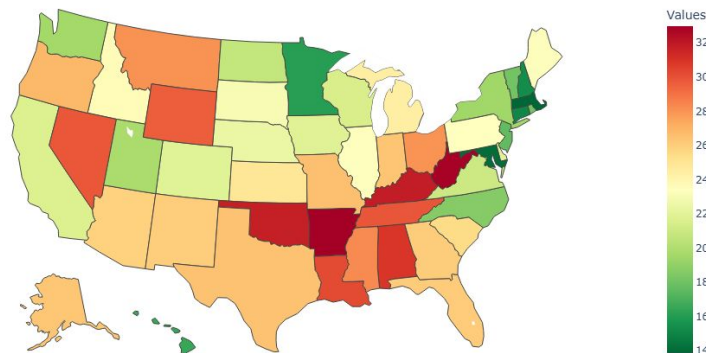


Fig. 1 U.S. States average health ranking for 2024 across all available measures

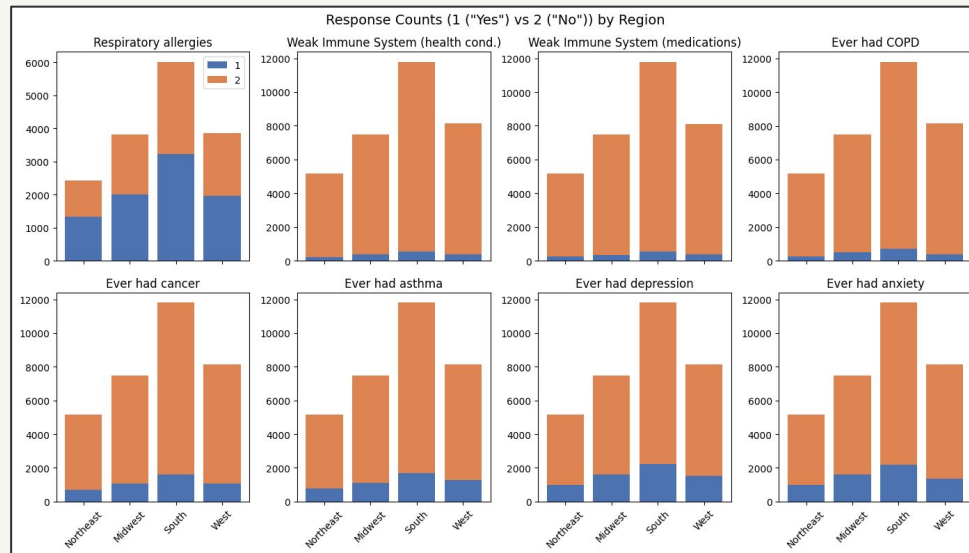


Fig. 2 Total number of adults across all four regions with pertinent medical diagnoses

Health Aspect: Next Steps

- Continued modification of the AHR dataset (including filtering unnecessary measures)
 - Will be informed by emerging research publications regarding various forms of pollution as well as findings from our environmental data, much like the NHIS data
 - Averaging state rank across demographics for each selected measure to create a singular comprehensive score
- Further investigation of selected measures for NHIS data
 - Determine which states fall under the umbrella of each region
 - Basic statistical analysis to determine if there is any significant difference(s) between regions for our measures
- Identify optimal communication strategies for the data
 - How will we present this information cohesively between the state and region level?

Political Aspect: Inspection

- Political datasets encode geography with FIPS code but data center data uses ZIP code, which required developing strategy to convert
 - Using Row Zero dataset of ZIP codes to County FIPS codes
 - Challenges: ZIPs across multiple FIPS, no matches from ZIP (match by county + state name)

	fips	data_center_count	REG_VOTERS	VOTER_TURNOUT_PCT	REG_VOTER_TURNOUT_PCT	PARTISAN_INDEX_DEM	PARTISAN_INDEX_REP
0	1001.0	0.0	43488.0	0.397022	0.410780	0.254452	0.745548
1	1003.0	0.0	189028.0	0.400008	0.389419	0.203950	0.796050
2	1005.0	0.0	17270.0	0.333925	0.381413	0.437367	0.562633
3	1007.0	0.0	14741.0	0.336907	0.400787	0.196562	0.803438
4	1009.0	0.0	41794.0	0.366511	0.399124	0.089329	0.910671

National Neighborhood Data Archive (NaNDA): Voter turnout and partisanship by county

- Filter to most recent year: 2022
- 3111 rows after merge

	fips	data_center_count	geo_name	state_abb	office_consolidated	year	winner	incumbent	prob_democrat	prob_republican	pid_est
0	1001.0	0.0	autauga	AL	Sheriff	2018.0	win	NaN	0.0	1.0	R
1	1003.0	0.0	baldwin	AL	Sheriff	2018.0	win	1.0	0.0	1.0	R
2	1003.0	0.0	baldwin	AL	County Legislature	2018.0	win	0.0	0.0	1.0	R
3	1003.0	0.0	baldwin	AL	County Legislature	2018.0	win	1.0	0.0	1.0	R
4	1003.0	0.0	baldwin	AL	County Legislature	2018.0	win	0.0	0.0	1.0	R

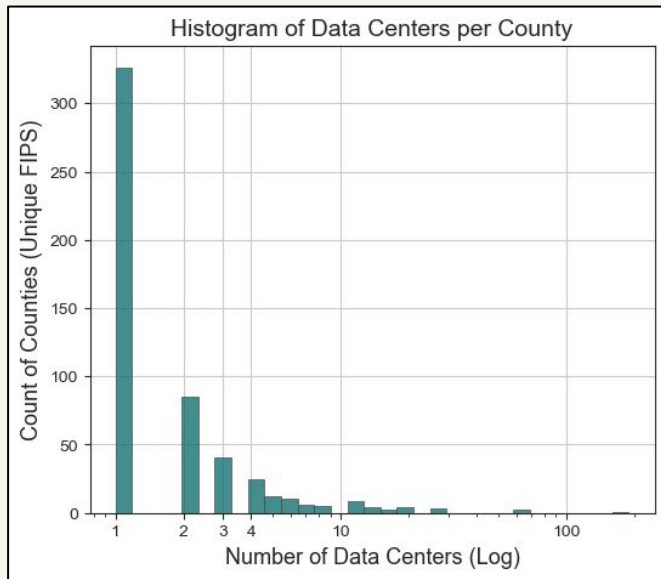
American Local Government Elections Database: Local election results from 1989-2021

- Filter to winners only
- 62,784 rows after merge

Political Aspect: EDA

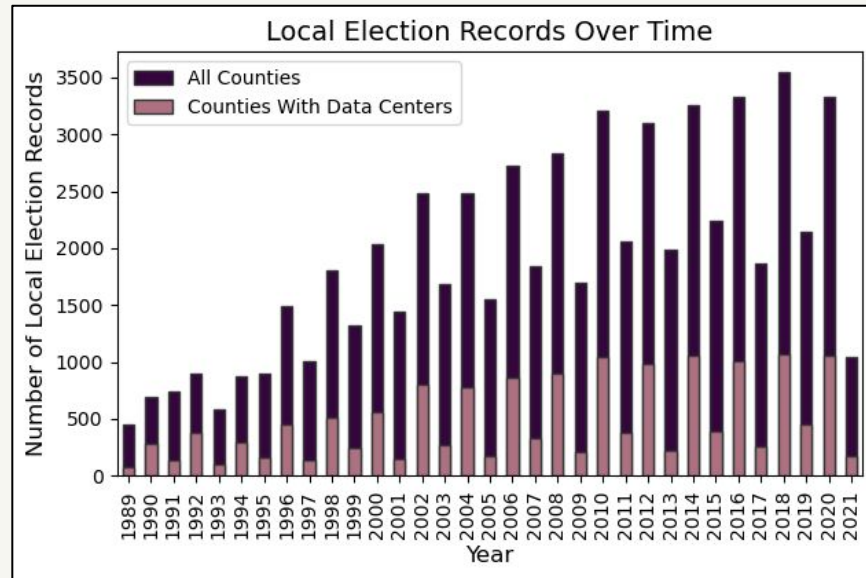
Result of ZIP to FIPS

For counties with at least one data center



American Local Government Elections Database

Considerations for aggregating election results over time



Political Aspect: Next Steps

- Aggregate additional FIPS-level information about data centers
- National Neighborhood Data Archive:
 - Create violin plots showing distribution of voter turnout and partisan index across different bins of data center presence
- American Local Government Elections Database:
 - Implement strategy to aggregate election results per county over ~2016-2021
 - Create visualizations to explore connections between local government and data center presence
- MIT precinct-level data for 2024 federal elections:
 - Most recent but least indicative of local politics, so potentially use as supplemental

Summary of Next Steps

Data Center Impact Explorer

Created by Anna Grace, Jillian Kunze,
Andrea MacGregor, and Morgan Murphy
for Drexel University's DSCI CAPSTONE

Data centers are the backbone of the modern Internet and provide the necessary physical storage space to enable cloud storage, digital streaming, and artificial intelligence applications (Malone, 2026). It is the latter that has been causing a particularly explosive growth in the construction of data centers across the U.S. in recent years (Stansbury et al., 2025). Yet that growth comes at a cost, and a groundswell of community opposition has raised concerns about the impact of data center construction and operation at the local level (Jones, 2026). Data centers require significant amounts of water and energy, often from natural gas, and can cause noticeable noise pollution. The centers can have a somewhat positive impact on the local economy, but the creation of stable, long-term jobs at these sites is often low (Malone, 2026). While research regarding the long-term effects of data centers is still ongoing, we hope that this site can help explain how data centers may affect the communities they are built in

Scroll to see the data! (*parallax preferably)



Data Center Impact Explorer

Select a **state** or **region** to see how data centers may impact residents



State DD v

Region DD v

(will be drop down
menu of states)

(will be drop down
menu of regions)

Politics

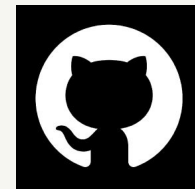
Environment

Power

Health

- Continue general refinement and other necessary tasks to make our data more usable downstream
- Location data inconsistency between datasets will likely be a major pain point to be considered
 - Region vs. state vs. ZIP/FIP code
- Identify optimal web page development and hosting platform(s)
 - How will we host and call our data?
 - How will we visualize our data on the web page?

GitHub Repository



github.com/agrace339/Data-Center-Impact-Explorer

- Folders to organize scripts and data
- Git Ignores
- First draft of ReadMe

The screenshot shows the GitHub repository page for 'Data-Center-Impact-Explorer' by user 'agrace339'. The repository is public and has 1 branch and 0 tags. The commit history shows a recent commit 'Remove .DS_Store' by 'd2e8edd' 7 minutes ago, with 6 commits in total. The file list includes 'Data', 'Script', '.gitignore', and 'README.md'. The README section is titled 'Data Center Impact Explorer' and describes a data science capstone project. The team section lists four members: Anna Grace, Jillian Kunze, Andrea MacGregor, and Morgan Murphy. The right sidebar shows repository statistics: 0 stars, 0 watching, and 0 forks. It also includes sections for Releases, Packages, Contributors (1), and Languages (Jupyter Notebook 100.0%).

Data-Center-Impact-Explorer Public

Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file Code

agrace339 Remove .DS_Store d2e8edd · 7 minutes ago 6 Commits

File	Commit	Time
Data	Initial commit	4 hours ago
Script	Remove .DS_Store	7 minutes ago
.gitignore	Update README.md	4 hours ago
README.md	Update README.md	4 hours ago

README

Data Center Impact Explorer

A data science capstone project investigating the environmental, infrastructure, political, and community impacts of data center development across the United States.

The project integrates multiple public datasets with the FracTracker Alliance Data Center Database to explore factors that may influence where data centers are built and how they affect surrounding communities.

Team

- Anna Grace
- Jillian Kunze
- Andrea MacGregor
- Morgan Murphy

About

No description, website, or topics provided.

Readme Activity 0 stars 0 watching 0 forks Report repository

Releases

No releases published [Create a new release](#)

Packages

No packages published [Publish your first package](#)

Contributors 1

agrace339 Anna Grace

Languages

Jupyter Notebook 100.0%

Dataset References

America's Health Rankings. (2025). America's Health Rankings 2025 annual report [Data set]. United Health Foundation.

<https://www.americashealthrankings.org/publications/reports/2025-annual-report>

Clary, W., Gomez-Lopez, I. N., Chenoweth, M., Gypin, L., Clarke, P., Noppert, G., Li, M., & Kollman, K. (2024). National Neighborhood Data Archive (NaNDA): Voter Registration, Turnout, and Partisanship by County, United States, 2004-2022 [Dataset]. Inter-university Consortium for Political and Social Research.

de Benedictis-Kessner, J., Lee, D. D. I., Velez, Y. R., & Warshaw, C. (2023). American local government elections database. *Scientific Data*, 10(1), 912.

National Health Interview Survey. (2024). 2024 NHIS Questionnaire, Datasets, and Documentation [Data set]. U.S. Centers for Disease Control and Prevention. <https://www.cdc.gov/nchs/nhis/documentation/2024-nhis.html>

Row Zero. (n.d.). FIPS codes for all U.S. locations in a spreadsheet. Retrieved July 21, 2026, from <https://rowzero.com/datasets/fips-codes-lookup#zip-code-mappings>

U.S. Drought Monitor. (n.d.). Data Tables. U.S. Drought Monitor. Retrieved June 30, 2026, from <https://droughtmonitor.unl.edu/DmData/DataTables.aspx>

U.S. Energy Information Administration (2025). EIA Bulk Data [Data set]. <https://www.eia.gov/opensdata/>



Questions?

GitHub Repository

<https://github.com/agrace339/Data-Center-Impact-Explorer#>

So easy to use, my cat can do it. (Pepper has a doctorate in github and advanced machine learning)

